

Lab 04

Topic

Insertion Sorting

Objective

To implement and analyze Insertion Sort algorithms by counting comparisons and swaps, comparing their behavior on different input cases.

Task (1):

Run Insertion Sort on:

1. A **sorted array**
2. A **reverse-sorted array**
3. A **random array**

Compare and display the **number of comparisons and swaps** for each case.

Task (2):

Between Bubble Sort, Selection Sort and Insertion Sort which algorithm is best for

1. A **sorted array**
2. A **reverse-sorted array**
3. A **random array**

Task (3):

Modify **Insertion Sort so that** the elements will move from left to right rather than right to left like it normally does. The array must be sorted in ascending order.